Efficiency vs. Robustness in Fiscal Shock Identification:

A Design-Based Sensitivity Analysis of Overidentified GMM Estimation

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Abstract

This project evaluates the performance of the overidentified GMM estimator in a fiscal Proxy SVAR framework. Using a set of sensitivity exercises, I examine how covariance restrictions and instrument quality affect the estimation of fiscal multipliers. The results show that overidentification improves statistical precision with little impact on point estimates, with gains that are larger for government spending shocks and under weaker instruments. In contrast, local projection estimates exhibit substantially wider confidence intervals, indicating a loss of precision. Overall, the findings highlight the efficiency and stabilizing role of GMM in environments with weak identification.

1 Introduction

The estimation of fiscal multipliers, the causal effects of government spending and tax changes on aggregate output, remains a central problem in empirical macroeconomics and a key input for policy design. Identification of these effects is challenging because fiscal policy is endogenous to macroeconomic conditions. Traditional structural vector autoregression (SVAR) approaches rely on timing assumptions or sign restrictions, whereas more recent contributions exploit external instruments, or proxy variables, to identify structural shocks without fully specifying the structural system.

Proxy-SVAR methods use external instruments that satisfy relevance and exclusion restrictions to identify individual shocks. However, when shocks are identified separately, the resulting system is typically just identified and does not exploit all the information contained in the covariance structure of the structural shocks. In particular, the assumption that structural shocks are mutually orthogonal is usually maintained but not imposed in estimation.

Gregory, McNeil, and Smith (2022) address this limitation by proposing an overidentified GMM estimator that combines instrumental-variable moment conditions with covariance restrictions on the structural shocks. In their framework, these orthogonality restrictions generate additional moments, allowing the model to be overidentified and enabling a formal J-test of the joint restrictions. In their fiscal application, this approach improves precision and reduces the standard errors of impact estimates.

This project evaluates whether imposing covariance restrictions through overidentified GMM improves the estimation of fiscal multipliers in a Proxy-SVAR framework. The focal metrics are impact fiscal multipliers and cumulative multipliers at medium horizons. The analysis adopts a structured sensitivity approach across alternative specifications, and can be interpreted as a meta-analytic comparison of estimates across designs rather than across studies. Rather than focusing on a single specification, the analysis examines how estimated multipliers and their precision vary across a set of modeling choices. In particular, it studies how the gains from overidentification depend on instrument quality, the type of fiscal shock (government spending versus tax revenue), and alternative specifications that do not impose the full SVAR structure.

The main objective is to characterize the conditions under which overidentification delivers reliable efficiency gains. While the GMM estimator is expected to improve precision, its performance may depend on the strength of the instruments and on the stability of the underlying economic environment. Accordingly, this study evaluates whether the benefits of imposing covariance restrictions are robust across different empirical settings, including comparisons with alternative estimation approaches.

The analysis makes two main contributions. First, it provides a quantitative assessment of the efficiency gains associated with overidentification in a fiscal SVAR setting. Second, it shows that these gains are not uniform: they are larger for spending shocks and increase when instruments weaken, highlighting an additional stabilizing role of covariance restrictions under weak identification, particularly relative to alternative approaches such as local projections.

The remainder of the study is organized as follows. Section 2 presents the model specification. Section 3 describes the data. Section 4 outlines the empirical design. Section 5 reports the results, and Section 6 concludes.

2 Model Specification

2.1 Structural Representation and External Instrument Identification

This section presents the econometric model used to estimate the focal fiscal multipliers, following the Proxy-SVAR framework in Gregory et al. (2022).

Let x_t denote an $N \times 1$ vector of endogenous variables, including government spending, tax revenue, and output. The reduced-form VAR of order p is given by:

$$x_t = \sum_{i=1}^{p} B_i x_{t-i} + D w_t + u_t, \quad u_t \sim IID(0, \Sigma), \tag{1}$$

where w_t collects deterministic components. The reduced-form residuals admit the structural representation:

$$u_t = \Theta \varepsilon_t, \quad \varepsilon_t \sim IID(0, \Omega), \tag{2}$$

where Θ is a nonsingular $N \times N$ matrix and Ω is diagonal, reflecting the orthogonality of structural shocks.

Let $z_{n,t}$ denote an external instrument (proxy variable) for the structural shock $\varepsilon_{n,t}$. The validity of the instrument requires the following moment conditions:

$$E[z_{n,t} \varepsilon_{n,t}] = \alpha_n \neq 0 \quad \text{(relevance)} \tag{3}$$

$$E[z_{n,t} \varepsilon_{m,t}] = 0, \quad m \neq n \quad \text{(exclusion)}. \tag{4}$$

Under these conditions, each instrument identifies a column of Θ. With one valid instrument per structural shock, the off-diagonal elements of Θ are exactly identified via instrumental variables estimation, yielding the standard proxy-SVAR estimator.

2.2 Overidentification via Covariance Restrictions and GMM

Although the just-identified IV estimator recovers Θ, the resulting estimated shocks,

$$\hat{\varepsilon}_t = \hat{\Theta}^{-1} u_t, \tag{5}$$

may exhibit non-zero sample correlations. This feature contradicts the maintained assumption that structural shocks are mutually orthogonal.

To incorporate this assumption directly into estimation, impose the covariance restrictions:

$$E[\varepsilon_{m,t}\varepsilon_{n,t}] = 0, \quad m \neq n. \tag{6}$$

These restrictions provide $\frac{N(N-1)}{2}$ additional moment conditions, which can be combined with the IV moment conditions to form an overidentified system. Let θ collect the unknown off-diagonal elements of Θ. The resulting estimator is obtained via Generalized Method of Moments (GMM), exploiting both:

1. Moment conditions from external instruments:

$$E\left[(u_{m,t} - \Theta_{m,n} u_{n,t}) z_{n,t}\right] = 0, \quad m \neq n, \tag{7}$$

which identify the columns of Θ;

2. Covariance restrictions implied by orthogonality of structural shocks:

$$E[\varepsilon_{m,t}\varepsilon_{n,t}] = 0, \quad m \neq n, \tag{8}$$

where $\varepsilon_t = \Theta^{-1} u_t$.

When these conditions exceed the number of unknown parameters, the model is overidentified. Estimation proceeds by minimizing the quadratic form:

$$J(\theta) = \bar{g}(\theta)' W \bar{g}(\theta), \tag{9}$$

where $\bar{g}(\theta)$ denotes the sample analog of the stacked moment conditions and W is a positive definite weighting matrix.

Estimation is implemented using an iterated GMM procedure. An identity weighting matrix is used to obtain initial estimates, followed by a heteroskedasticity and autocorrelation consistent (HAC) weighting matrix based on the Newey–West (1987) estimator with four lags. The resulting nonlinear optimization problem is solved using a BFGS algorithm, following Gregory et al. (2022).

First, the additional moment conditions improve efficiency: in the fiscal application of Gregory et al. (2022), the standard errors of the impact coefficients decline by approximately 30%. Second, overidentification allows for specification testing through Hansen's J-test, offering a formal evaluation of the joint validity of the instruments and covariance restrictions.

Importantly, the covariance restrictions can also contribute to identification. In particular, when an instrument for one structural shock is unavailable, the combination of IV and covariance moments may still suffice to identify the full matrix Θ, a feature that is especially valuable in macroeconomic applications where valid instruments are scarce.

The parameters estimated in the matrix Θ directly map into the impulse response functions used to construct the focal metrics of this study, namely impact and cumulative fiscal multipliers. In particular, the output response to identified fiscal shocks forms the basis for computing these multipliers across horizons.

To assess the robustness of these results to alternative specifications, the analysis also considers an estimation approach based on local projections, which is introduced in the empirical design section.

3 Data

3.1 Data Sources and Baseline Sample

The empirical analysis follows Gregory et al. (2022) and uses quarterly U.S. data from 1948:Q1 to 2019:Q4. The reduced form VAR includes real federal government spending (g_t), real federal tax revenue (τ_t), and real GDP (y_t), all expressed in per capita logarithms and deflated by the GDP deflator.

The variables are constructed to match the definitions in Gregory et al. (2022). Government spending (g_t) corresponds to federal government expenditure and gross investment, tax revenue (τ_t) includes federal current tax receipts, social insurance contributions, and corporate income taxes, and output (y_t) is measured as real GDP. All series are obtained from the Bureau of Economic Analysis (BEA), expressed in per capita terms, deflated using the GDP deflator, and transformed into natural logarithms.

The external instruments replicate those used in the original study. Government spending shocks are instrumented using the military spending news series of Ramey and Zubairy (2018), which captures revisions in the expected present value of government purchases associated with military buildups. Tax shocks are instrumented using the narrative tax series of Mertens and Montiel Olea (2018), based on exogenous tax changes. Output shocks are instrumented using innovations to total factor productivity (TFP) from the Federal Reserve Bank of New York (FRBNY) DSGE model, where TFP captures exogenous changes in overall production efficiency not explained by inputs.

The underlying VAR specification also follows Gregory et al. (2022). It is estimated with four lags, a quadratic time trend, and dummy variables for 1975Q2 and its lags.

3.2 Summary Statistics

Table 1 reports summary statistics for the main variables used in the analysis. Output, government spending, and tax revenue exhibit moderate dispersion over the sample, with output being the least volatile and tax revenue displaying comparatively higher variability.

The external instruments display markedly different distributional properties. The government spending instrument shows substantial variation and pronounced extreme values, reflecting the episodic nature of military spending news shocks. In contrast, the tax instrument is tightly distributed around zero, while the TFP-based instrument exhibits relatively low variance by construction. These differences are relevant for instrument strength and motivate the sensitivity analysis conducted below.

Table 1: Summary Statistics

Variable	Mean	Std. Dev.	Min	Max
Real GDP (log, per capita)	3.765	0.365	3.003	4.302
Government Spending (log, per capita)	1.435	0.181	0.666	1.745
Tax Revenue (log, per capita)	1.967	0.379	0.969	2.486
Spending Shock Instrument	0.441	4.573	-8.913	61.780
Tax Shock Instrument	-0.029	0.227	-2.110	0.220
TFP Shock Instrument	0.000	0.100	-0.340	0.484
Short Term Interest Rate	4.039	3.064	0.013	15.053

4 Analysis Design

4.1 Baseline Replication

The empirical strategy begins by replicating the baseline results in Gregory et al. (2022) using both a just identified Proxy SVAR estimated via instrumental variables and an overidentified GMM estimator that incorporates covariance restrictions on the structural shocks. This step ensures that the implementation matches the main findings in the original study before introducing additional variation.

Each specification is treated as a distinct design point, and the collection of estimates across specifications forms a meta analysis dataset. This approach allows for a systematic comparison of how estimated outcomes vary with key modeling choices.

4.2 Sensitivity Design

Building on the baseline, the analysis examines how key estimated objects vary across alternative specifications. Each estimated outcome, such as an impact coefficient or a fiscal multiplier at a given horizon, is treated as a unit of comparison. The framework introduces systematic variation in instrument quality and alternative specifications in order to evaluate the robustness of both point estimates and statistical precision.

One source of variation comes from instrument quality. Starting from the baseline specification, two perturbations are considered. In one case, instruments are set to missing prior to a given cutoff date, reducing the effective sample available for estimation. In another, random noise is added to the instruments to weaken their informational content. These modifications provide a controlled way to assess how weaker or less informative proxies affect estimation and precision.

To implement the instrument quality perturbations, I construct two alternative designs. In the trimmed specification, I retain the original instrument series but set all observations prior to a fixed cutoff date, 1980Q1, to missing values. This reduces the effective sample available for estimation without altering the underlying VAR sample.

In the noisy specification, I contaminate each instrument by adding mean zero Gaussian noise scaled to its standard deviation. Specifically, each proxy is replaced by the sum of the original series and a noise component proportional to 0.35 times its sample standard deviation. This procedure lowers the signal to noise ratio of the instruments while preserving their overall scale. Both perturbations are applied symmetrically to all three instruments, government spending, tax, and output shocks, ensuring a consistent comparison across specifications.

In addition, the analysis incorporates an alternative estimation approach based on local projections following Jordà (2005), estimated with instrumental variables. This method estimates impulse responses directly through a sequence of horizon-specific regressions, rather than imposing a full VAR structure. In each horizon, the outcome variable is regressed on the identified shock using external instruments, allowing for flexible dynamics without requiring a global system specification. While this approach is generally more robust to misspecification, it may be less statistically efficient than VAR-based methods, as it does not exploit cross-equation restrictions or covariance conditions.

Finally, each specification is estimated using both instrumental variables and overidentified GMM. In addition, the local projection specification is estimated using instrumental variables. The comparison focuses on impact effects, cumulative fiscal multipliers, and associated measures of statistical precision, providing a consistent way to evaluate how overidentification affects both magnitudes and uncertainty.

4.3 Comparative Metrics

To summarize the effects of these variations, the analysis focuses on a set of key statistics that map directly into the main claims in Gregory et al. (2022). The comparison begins with the estimated impact coefficients in the matrix Θ, with particular attention to the response of output to government spending and tax shocks. These estimates provide a direct benchmark for assessing whether overidentification alters the magnitude of the effects.

In parallel, the analysis evaluates statistical precision using standard errors and confidence interval widths. The central object of interest is the relative efficiency gain from imposing covariance restrictions, which captures the extent to which GMM improves precision relative to IV.

The analysis also considers the correlation structure of the estimated structural shocks. Because the IV estimator does not impose orthogonality, the resulting shocks may exhibit non zero correlations. By contrast, the GMM estimator incorporates covariance restrictions as additional moment conditions, leading to shocks that are closer to orthogonal. Examining these patterns provides insight into how each estimator aligns with the underlying structural assumptions.

To connect the results to policy relevant objects, fiscal multipliers are compared across specifications, with particular emphasis on cumulative multipliers at medium horizons, such as $h = 10$ quarters. These measures summarize the dynamic effects of fiscal shocks in a way that is directly comparable to the existing literature.

Finally, efficiency gains from GMM are summarized through relative reductions in standard errors, defined as

$$\text{Gain} = \frac{SE^{IV} - SE^{GMM}}{SE^{IV}}. \tag{10}$$

These gains provide a direct empirical counterpart to the central claim that overidentification improves the precision of estimated multipliers.

5 Results

5.1 Baseline Results and Replication

The baseline specification closely replicates the main findings of Gregory et al. (2022). Estimated impact responses of output to fiscal shocks are reported in Table 2. Point estimates are very similar across IV and GMM, indicating that overidentification has little effect on magnitudes. In contrast, precision improves markedly under GMM: the standard error for government spending shocks declines by approximately 35%, while the reduction for tax shocks is around 18%.

Table 2: Baseline Estimates: Output Response to Fiscal Shocks

	IV	GMM	SE (IV)	SE (GMM)
$Y \leftarrow G$	0.206	0.191	0.119	0.078
$Y \leftarrow \tau$	-0.052	-0.054	0.070	0.057

A similar pattern emerges for fiscal multipliers (Table 3). While point estimates remain stable, standard errors are substantially smaller under GMM, reinforcing the interpretation that covariance restrictions primarily enhance efficiency.

Table 3: Baseline Multipliers (Impact, $h = 0$)

	IV	GMM	SE (IV)	SE (GMM)
Spending	2.03	1.88	1.18	0.77
Taxes	-0.31	-0.33	0.42	0.34

Taken together, these results confirm that overidentification improves statistical precision without materially affecting point estimates.

5.2 Instrument Quality and Identification

To evaluate robustness, instrument quality is varied using trimmed and noisy specifications. First-stage statistics are summarized in Table 4. The trimmed design substantially weakens the output instrument, with $F_Y \approx 2$, indicating a potential loss of instrument strength.

Table 4: First-Stage Strength (F-statistics)

	F_G	F_T	F_Y
Baseline	9.94	24.38	10.56
Trimmed	10.23	16.29	1.99

Under this weaker instrument environment, the results in Table 5 show that efficiency gains from GMM persist for spending shocks but largely disappear for tax shocks.

Table 5: Trimmed Instruments

	IV	GMM	SE (IV)	SE (GMM)
$Y \leftarrow G$	0.156	0.173	0.125	0.089
$Y \leftarrow \tau$	-0.191	-0.192	0.051	0.050

The contrast between estimators becomes more pronounced under severe noise (Table 6). In this case, IV estimates become highly unstable, whereas GMM remains well-behaved. The implied spending multiplier rises sharply under IV but remains moderate under GMM, highlighting the stabilizing role of covariance restrictions when instruments are weak or contaminated by noise.

Table 6: Noisy Instruments

	IV	GMM	SE (IV)	SE (GMM)
$Y \leftarrow G$	0.662	0.244	0.454	0.106
$Y \leftarrow \tau$	-0.046	-0.042	0.072	0.067

5.3 Shock Properties and Overidentification

An important advantage of the GMM estimator is its ability to impose approximate orthogonality across structural shocks. The correlation patterns reported in Table 7 illustrate this feature. While IV estimation yields sizable correlations, particularly under noisy instruments, GMM substantially attenuates these relationships, producing shocks that are more consistent with the underlying structural assumptions.

Table 7: Correlation Between Estimated Shocks

	$\rho(G,T)$	$\rho(G,Y)$	$\rho(T,Y)$
IV (Baseline)	0.164	-0.132	0.020
IV (Noisy)	0.500	-0.678	-0.260
GMM (Baseline)	0.150	-0.078	0.042
GMM (Noisy)	0.136	-0.061	0.013

Formal evidence from the J-test further supports the validity of these restrictions (Table 8). Despite some decline in p-values under noisy conditions, the overidentifying restrictions are not rejected in any specification.

Table 8: J-Test Results

	J-stat	p-value
Baseline	0.031	0.999
Trimmed	0.813	0.846
Noisy	2.441	0.486

5.4 Alternative Specification: Local Projections

To further assess the robustness of the results, I consider an alternative specification based on local projections estimated with instrumental variables. This approach does not impose the full VAR structure or the covariance restrictions used in the overidentified GMM estimator, and therefore serves as a useful benchmark for evaluating the role of additional identifying restrictions.

Figure 1 compares the cumulative spending multiplier at horizon $h = 10$ across the three specifications. The SVAR-IV estimate is positive and moderately precise, while the SVAR-GMM estimate remains similar in magnitude but is estimated more precisely. By contrast, the local projection estimate is much less stable: the point estimate turns negative and the confidence interval becomes extremely wide, clearly crossing zero. This reflects a substantial loss of statistical precision relative to the SVAR-based estimates.

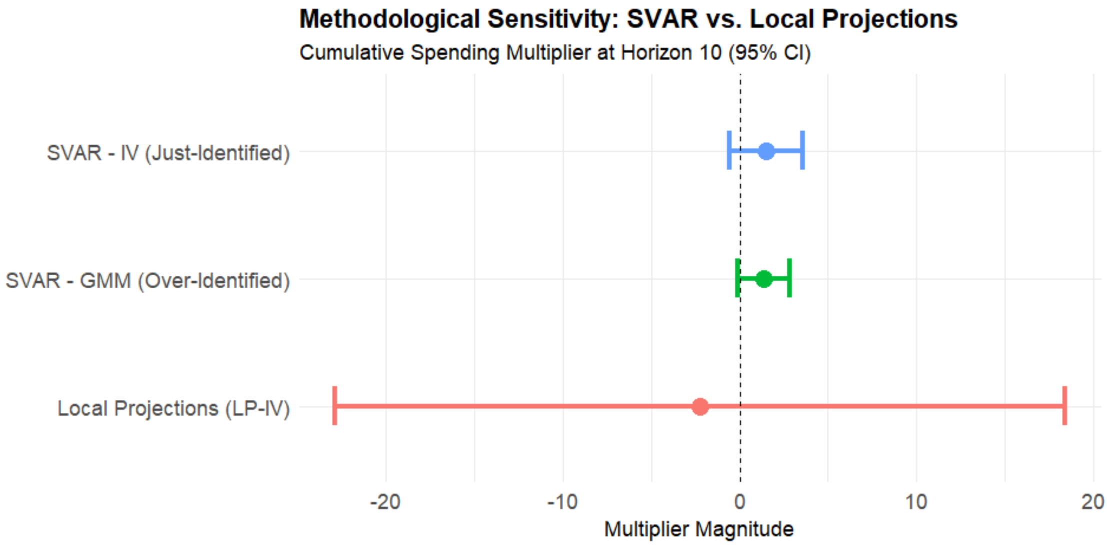

Figure 1: Methodological Sensitivity: SVAR vs. Local Projections. The figure reports cumulative spending multipliers at horizon $h = 10$ with 95% confidence intervals.

Table 9 summarizes the same comparison numerically. The SVAR-IV estimate equals 1.4967 with a standard error of 1.0559, while the overidentified GMM estimate is 1.3733 with a standard error of 0.7382. This implies a relative efficiency gain of GMM relative to IV of 1.4303, measured as the ratio of the IV standard error to the GMM standard error. The local projection estimate is -2.2358 with a standard error of 10.5282, corresponding to a relative efficiency loss of 14.2612 relative to GMM.

Table 9: Cumulative Spending Multiplier at Horizon 10

Method	Multiplier	Std. Error	Rel. Eff.
SVAR-IV (Just-Identified)	1.4967	1.0559	1.4303
SVAR-GMM (Over-Identified)	1.3733	0.7382	1.0000
Local Projections (LP-IV)	-2.2358	10.5282	14.2612

Overall, the comparison suggests that local projections provide a flexible robustness check, but at the cost of much weaker precision in this application. The overidentified GMM estimator preserves the qualitative conclusion of a positive spending multiplier while delivering substantially tighter inference. This reinforces the main result of the paper: covariance restrictions improve efficiency and stabilize estimates, especially in settings where identification is fragile.

5.5 Efficiency Gains Across Specifications

A concise summary of efficiency gains is reported in Table 10. Reductions in standard errors are consistently larger for spending shocks than for tax shocks and increase markedly under noisy instruments. This pattern indicates that covariance restrictions are particularly valuable in environments with weaker identification.

Table 10: Relative Efficiency Gains from GMM

	Baseline	Trimmed	Noisy
Spending (g)	0.35	0.29	0.77
Taxes (τ)	0.18	0.01	0.08

5.6 Meta-Regression Evidence

To summarize the patterns observed across specifications, I estimate a simple meta regression that relates efficiency gains to key design features. Each specification shock pair is treated as an observation in a meta analytic dataset, where the dependent variable is the relative reduction in standard errors achieved by the GMM estimator.

The meta regression is specified as:

$$Gain_i = \beta_0 + \beta_1 Tax_i + \beta_2 Noisy_i + \beta_3 Trimmed_i + \varepsilon_i \tag{11}$$

where $Gain_i$ denotes the efficiency gain for observation i, Tax_i is an indicator for tax shocks, and $Noisy_i$ and $Trimmed_i$ are indicators for the alternative instrument designs.

This specification allows for a direct interpretation of how design features affect the efficiency of estimation. The coefficient β_1 captures systematic differences between tax and spending shocks, while β_2 and β_3 quantify how reductions in instrument quality alter the gains from overidentification. As such, the meta-regression serves as a formal summary of the sensitivity analysis, linking variation in empirical design to changes in estimator performance.

The estimated regression shows that efficiency gains are substantially smaller for tax shocks, with a coefficient of approximately -0.38 relative to government spending shocks. In contrast, gains increase under noisy instrument designs, with an estimated coefficient of about 0.15, indicating that covariance restrictions become more valuable when identification is weaker. The trimmed specification is associated with slightly lower gains, although this effect is modest.

These results suggest that overidentification is particularly valuable in empirical settings where instruments are weak or noisy, a common feature in fiscal applications.

Given the small number of observations, these results should be interpreted as descriptive rather than inferential. Nonetheless, they provide a useful summary of the patterns observed across specifications and reinforce the conclusion that the benefits of overidentification are larger for spending shocks and in weaker identification environments.

Economically, these results imply that the benefits of imposing additional identifying restrictions are not uniform, but depend critically on the empirical environment. In settings where instruments are weak or noisy, the use of overidentifying restrictions can substantially improve the reliability of inference, reinforcing their practical relevance in applied macroeconomic work.

6 Conclusion

This project evaluates the role of overidentification via covariance restrictions in a Proxy-SVAR framework, following Gregory et al. (2022), and examines its performance across a range of empirical environments. Using a structured set of sensitivity exercises, the analysis compares the standard just-identified IV approach with an overidentified GMM estimator that incorporates additional moment conditions implied by the orthogonality of structural shocks. In addition, the analysis considers an alternative specification based on local projections estimated with instrumental variables as a complementary benchmark.

The results show that overidentification consistently improves statistical precision while leaving point estimates largely unchanged. In the baseline specification, the GMM estimator delivers substantially smaller standard errors, particularly for government spending shocks, closely replicating the findings in the existing literature.

Beyond the baseline, the analysis highlights important heterogeneity in these gains. Improvements in precision are systematically larger for spending shocks than for tax shocks and become more pronounced as instrument quality deteriorates. In weaker identification environments, GMM not only enhances efficiency but also stabilizes estimates relative to IV, reducing the sensitivity of results to noise and sample variation.

The comparison with local projections further reinforces these findings. While the alternative specification yields broadly similar point estimates, it is associated with substantially wider confidence intervals, reflecting a loss of statistical precision in the absence of covariance restrictions. This contrast highlights the role of overidentification as a mechanism to obtain more informative inference in empirical settings with limited or noisy instruments.

From a policy perspective, these findings suggest that fiscal multiplier estimates based on weak instruments should be interpreted with caution unless additional identifying restrictions are imposed. In such settings, overidentified approaches can provide more reliable inference by reducing estimation uncertainty and improving the stability of estimated effects.

Future research could extend this framework to alternative identification strategies, such as sign-restricted or narrative SVARs, and to richer macroeconomic environments. Exploring the performance of overidentified methods in nonlinear or state-dependent settings may further clarify their robustness and practical relevance in applied work.

References

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